Seeing Your Speech Style: A Novel Zero-Shot Identity-Disentanglement Face-based Voice Conversion

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Face-based Voice Conversion

- Voice Conversion (VC) aims to change the speaker identity in speech from a source speak er to that of a target speaker, while preserving the linguistic content.

- Zero-Shot Face-based Voice Conversion is explored.

- Application
  - Automated film dubbing
  - Personalized virtual assistants(e.g., siri, bixby)

Challenge

- **The fundamental challenge** is to accurately map identity information between faces and voices.

1. **Acquisition of facial embeddings that are well-aligned with the speaker's voice identity**
   - Now, focus on general facial features rather than speaker-specific features(FVMVC, FaceVC)
   - ➢ the models become highly dependent on the training data and lack the ability to locate unique voice character istics among different speakers

2. **Decoupling of content and speaker identity information from the audio input**
   - FVMVC use mixed supervision strategy
   - However, in practical scenarios, it is often difficult to acquire adequate and balanced supervision
   - ➢ Lead to suboptimal decoupling performance

To Address Challenge 1

- We design **an Identity-Aware Query-based Contrastive Learning (IAQ-CL) module**
  - Instead of adopting static encoding methods
  - Precisely extract the most identity-relevant facial features.

- **Identity-Aware Query-based Contrastive Learning (IAQ-CL) Module**
  - **Self-Adaptive Face-Prompted QFormer (SAFPQ)**
    - Employs learnable self-adaptive face prompts.
    - Queries identity-relevant features from a frozen CLIP visual encoder (Radford et al., 2021).
  - **Contrastive Learning** with identity features extracted from audio

To Address Challenge 2

- We propose a novel **Mutual Information-based Dual Decoupling (MIDD) module**
  - Enhance feature purification by addressing overlapping information in speech features.

- **Mutual Information-based Dual Decoupling (MIDD)**
  - **Speech Feature Decoupling**
    - Decomposes speech into subspaces representing different attributes.
    - Minimizes overlap between the speaker identity and content features by Mutual Information (MI).
  - **Fine-Grained Speaker Supervision**
    - Emphasizes subtle speaker distinctions.
    - Prevents model collapse by leveraging speaker identity.

Text-Based Speech Generation with Style Control

- **Additional Problem with Previous Approaches**
  - **Voice Input-Based Models**
    - Reference audio is not always available in practical scenarios.
    - e.g., digital humans, historical figures
  - **Text Input-Based Models**
    - Lack the flexibility to manipulate speech style
    - Often produce speech in a "machine" manner

- **Contribution**
  - Incorporate **text as an alternative modality** during the inference stage.
  - Propose **a style-controllable strategy** to control over **emotion** and **speed of speech**.
  - ➢ Enables the **generation of natural, rhythmical, and controllable speech** from text.

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Method

- ID-FaceVC employs an **end-to-end training** approach.

- **Main Components**

  1. **ID-Aware Query-based Contrastive Learning Module**
     - Aligns facial features with voice identity.
  2. **Mutual Information-based Dual Decoupling Module**
     - Separates content and speaker identity in audio input.
  3. **Alternative Text-Input with Style Control Module**
     - Allows text-based input with emotion and speed control.

Architecture of the proposed ID-FaceVC

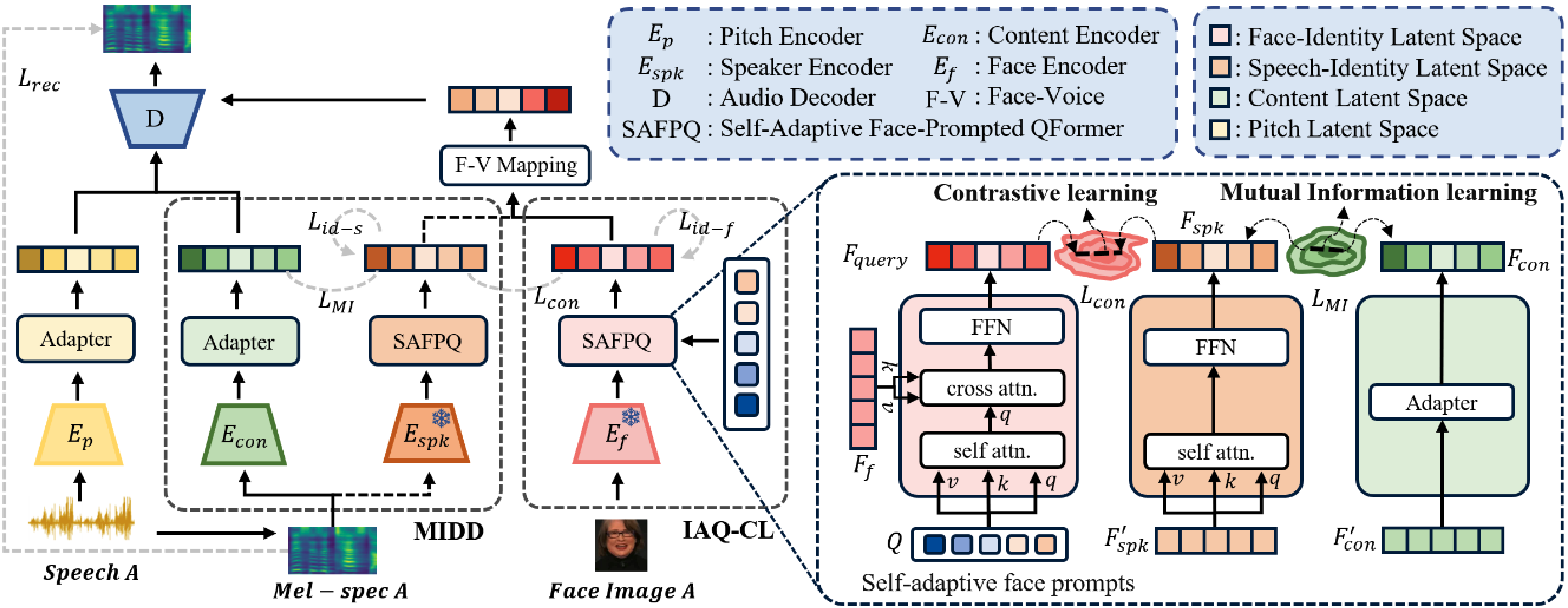

Figure 2: Overview of the proposed ID-FaceVC. The Adapter is a Feed Forward Network used to adjust vector dimensions. The embeddings F_f, F'_{spk}, F'_{con} correspond to the face, speaker, and content features extracted by E_f, E_{spk}, E_{con}, respectively.

ID-Aware Query-based Contrastive Learning

- **Purpose**
  - Extract facial features aligned with the speaker's voice identity.

- **Key Components**
  1. **Self-Adaptive Face-Prompted QFormer**
     - Adapts facial features dynamically for better voice identity alignment.
  2. **Face-Related Speaker Identity Supervision**
     - Ensures accurate mapping between facial features and speaker identity.

- **Outcome**
  - Enhanced feature alignment for natural and accurate voice conversion.

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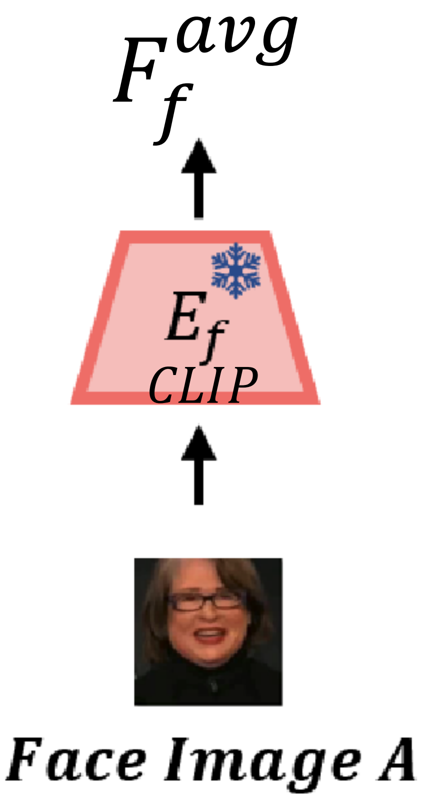
Using CLIP as Face Encoder

- **CNN-based Limitations**: Struggles with the diversity of facial features.

- **Approach**
  1. **Frozen CLIP Visual Encoder**
     - Uses a pretrained CLIP (Radford et al. 2021) model to extract robust facial features.
  2. **Frame-wise Visual Embedding Averaging**
     - Computes the arithmetic mean of embeddings across frames.
     - Avoids sampling bias by eliminating dependence on a single frame.

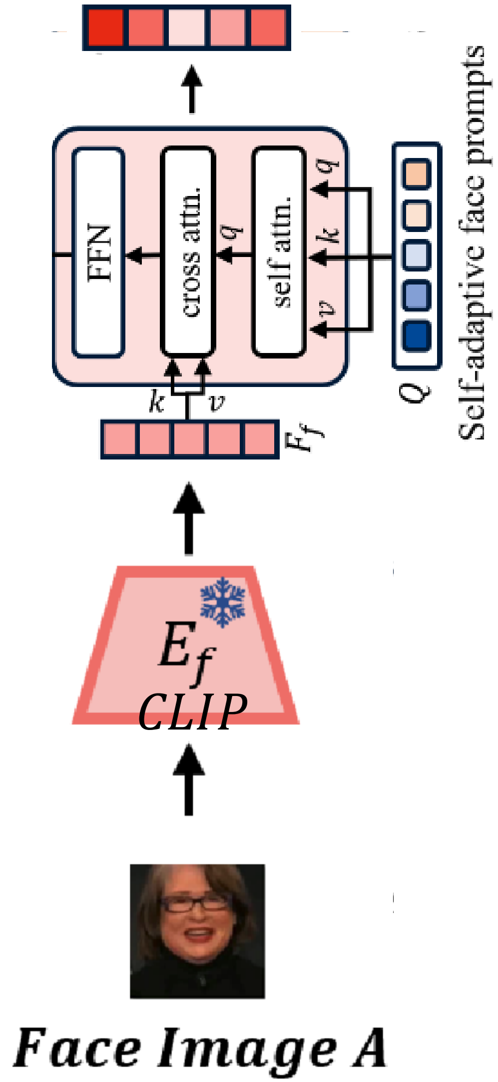


Self-Adaptive Face-Prompted QFormer

- **CLIP Visual Features**
  - High-dimensional and redundant.
  - Include **unnecessary information**.
    (e.g., facial expressions, head poses, backgrounds)
  - **Small portion** related to speaker style.

- **Self-Adaptive Face-Prompted QFormer (SAFPQ)**
  - Filters speech-relevant features from facial embeddings.
  - Acts as an information bottleneck, eliminating redundancy.
  - Enhances identity integration across face and voice domains.

*information bottleneck: process of selectively passing on the important information and reducing the unnecessary.



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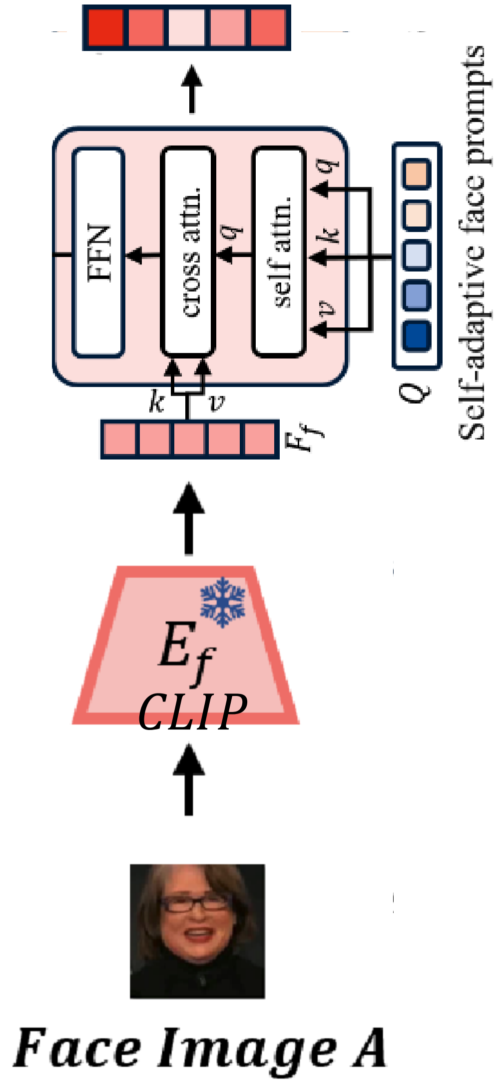
Self-Adaptive Face-Prompted QFormer

- **Key Features**
  1. **Enhanced Feature Filtering**
     - Extracts only identity-relevant features from facial embeddings.
  2. **Identity Cohesion**
     - Combines identity information from face and voice domains.

- **Inference Stage**
  - **Self-Adaptive Face Prompts (SAFP)**
    - Dynamically retrieve identity-relevant features.
    - Facilitate prediction of speaker's style from unseen facial images.



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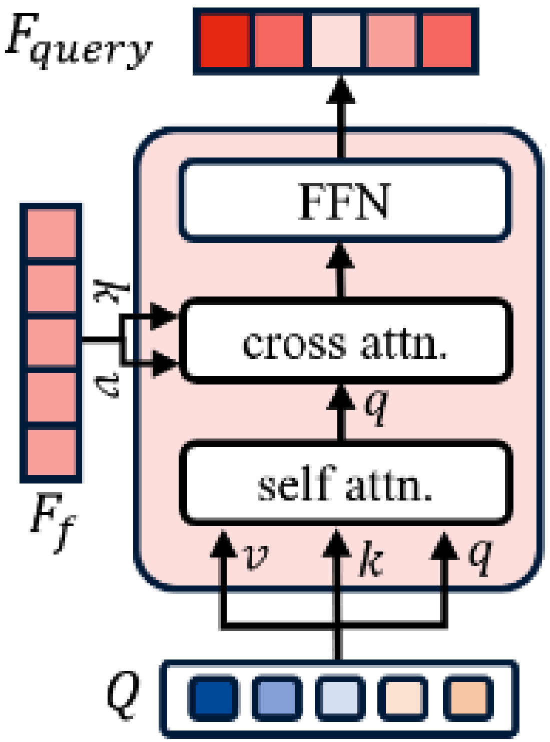
SAFPQ Workflow

- **Workflow Overview**
  1. **Initialize SAFP**: Learnable prompts are established and updated through self-attention.
  2. **Self-Attention**: The most informative prompts are highlighted from global perspective.
  3. **Cross-Attention**: SAFP retrieves identity-relevant features by interacting with facial embeddings.
  4. **Fully Connected Layers**: FFN fuses the retrieved features for final representation.

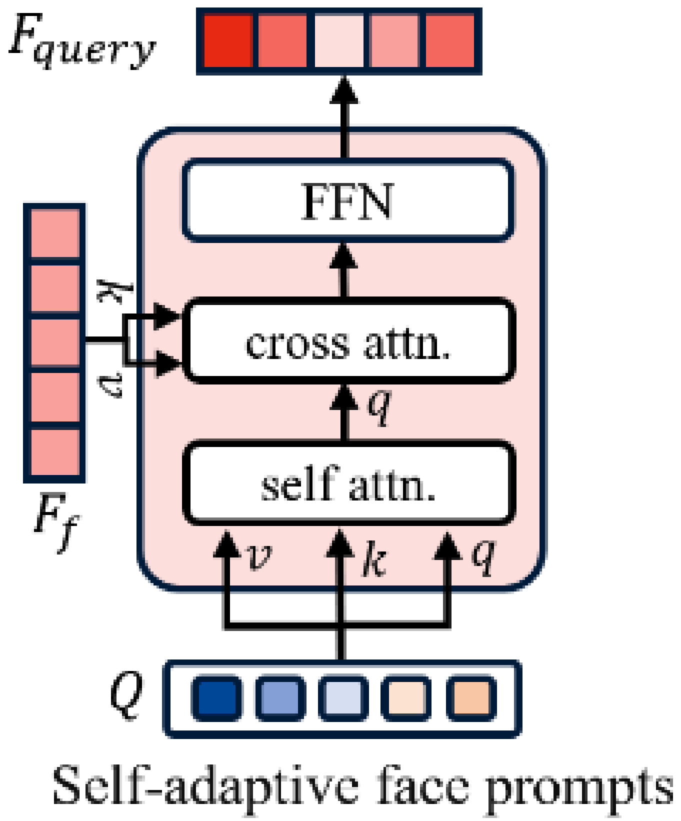


Self-adaptive face prompts

Mathematical Process of SAFPQ

- **Self-Attention Mechanism**

$$A_{self} = softmax\left(\frac{\boldsymbol{Q}W_q^{self}\left(\boldsymbol{Q}W_k^{self}\right)^T}{\sqrt{d_k}}\right)\boldsymbol{Q}W_v^{self}$$



- **Components**
  - $W_q^{self}, W_k^{self}, W_v^{self}$: The learnable weights for the self-attention
  - $\boldsymbol{Q}$: Self-Adaptive Face Prompts
  - d_k: Dimension of the key

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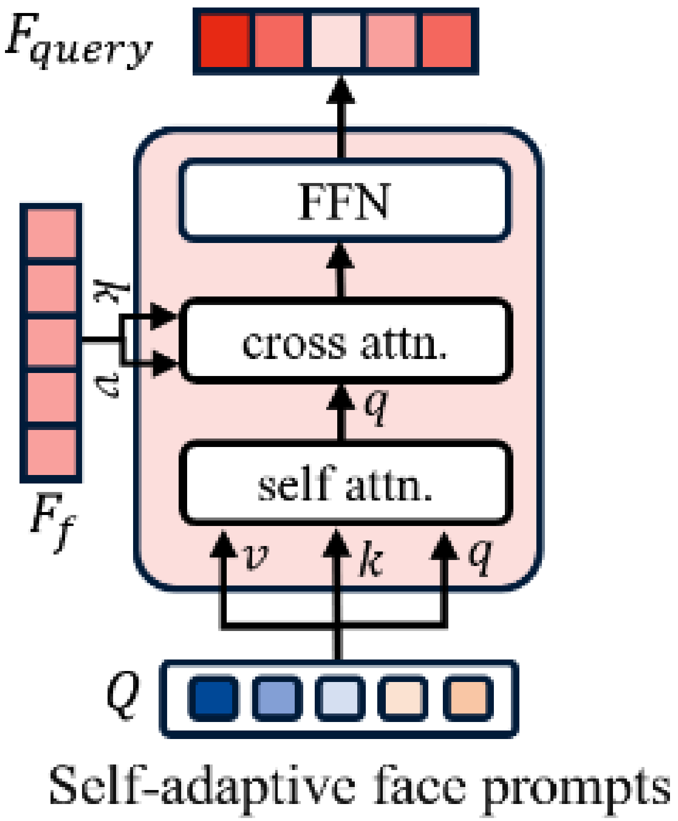
Mathematical Process of SAFPQ

- **Cross-Attention Mechanism**

$$A_{cross} = softmax\left(\frac{A_{self}W_q^{cross}\left(F_f W_k^{cross}\right)^T}{\sqrt{d_k}}\right)F_f W_v^{cross}$$



- **Components**
  - $W_q^{cross}, W_k^{cross}, W_v^{cross}$: The learnable weights for the self-attention
  - F_f: The facial embedding extracted by the CLIP

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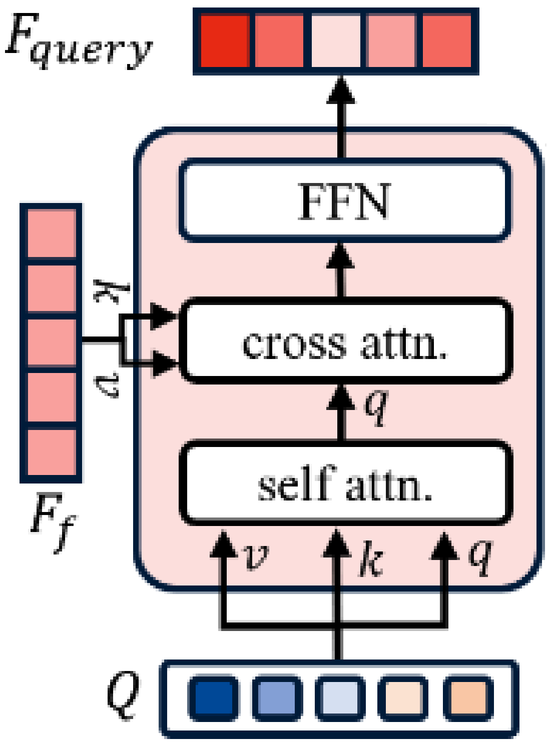
Mathematical Process of SAFPQ

- **Fully Connected Layer**

$$F_{query} = FFN(A_{cross})$$

- **Components**
  - F_{query}: The final queried facial features

➢ Extract facial features that are well-aligned with the speaker's voice identity



Self-adaptive face prompts

Loss of SAFPQ

- **Construction Loss**

$$L_{con} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N} y_{i,j} \log\left(\frac{\exp(sim(i,j)/\tau)}{\sum_{k=1}^{N}\exp(sim(i,k)/\tau)}\right)$$

- **Components**
  - N: The number of samples in a batch
  - τ: Temperature hyperparameter
  - i: fixed index for facial embeddings
  - j, k: speech embeddings
  - $y_{i,j}$: indicator function
    - If samples i and j belong to the same speaker, then $y_{i,j} = 1$; otherwise, $y_{i,j} = 0$.

Face-related Speaker-identity Supervision

- **Objective** Enhance the model's ability to differentiate facial features between speakers.

- **Mechanism**
  - **Consistency**: Ensures facial features remain stable for the **same speaker**.
  - **Diversity**: Promotes clear differentiation for **different speakers**.

- **Goal**
  - **Consistency**: Generate similar speech styles for **different images of the same speaker**.
  - **Diversity**: Create distinct speech styles for **different speakers**.

Face-related Speaker-identity Supervision

- **Loss**

$$L_{id-f} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{i=1}^{C} t_{ni} \cdot \log(p_{ni})$$

- **Components**
  - C: The number of distinct speakers
  - t_{ni}: The one-hot encoded target label for the n-th sample
  - p_{ni}: The softmax probabilities that the n-th sample's F_{query} belong to the i-th speaker

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Mutual Information-based Dual Decoupling (MIDD)

- **Purpose**
  - Precisely represent **disentangled latent spaces**.
  - Purify **speech content information**.

- **Key Components**
  - **Disentangled Latent Space**
  - **Mutual Information-based Decoupling**

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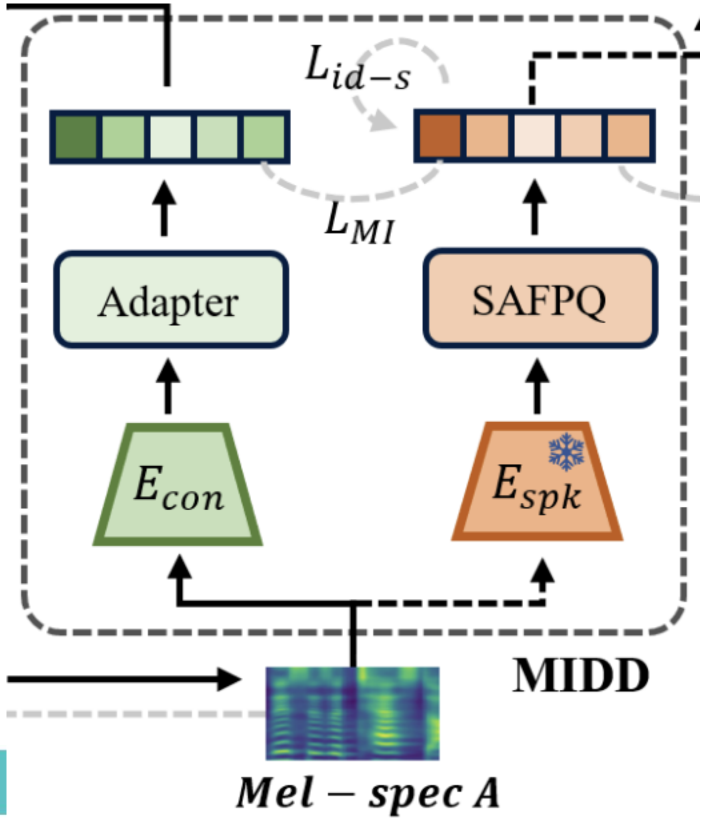
Disentangled Latent Space

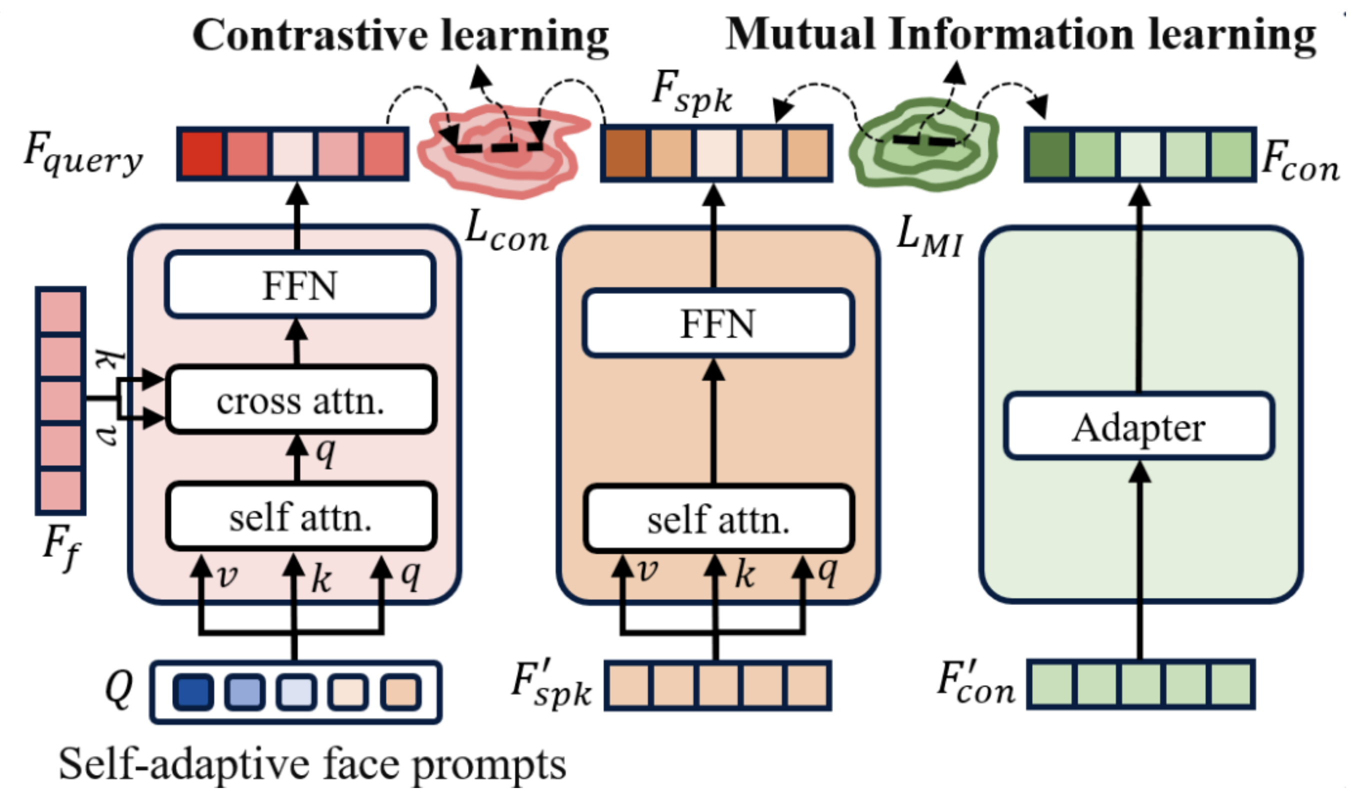
- **Objective**: Achieve robust content representation by **removing non-content-related features**.

- **Approach**
  - **Decomposition**: Split speech into subspaces for **speaker style (F_{spk})** and **content (F_{con})**.
  - **Speaker Style Encoder**
    - Utilizes **Contrastive Language-Audio Pretraining (CLAP)** (Wu et al., 2023).
    - Processes features through **SAFPQ** (similar to facial embedding handling).
  - **Content Encoder**
    - Combines **vector quantization** (Bitton et al., 2020) and **contrastive predictive coding** (Oord et al., 2018).
- **Tools Used**:
  - **CLAP** for speaker encoding.
  - **SAFPQ** for feature processing (without cross-attention).

Disentangled Latent Space

- **Tools Used**
  - **CLAP** for speaker encoding.
  - **SAFPQ** for feature processing (without cross-attention).





Mutual Information-based Decoupling

- **Challenge**
  - Simply separating latent spaces for speaker and content embeddings does not guarantee sufficient **feature decoupling**.
  - Previous methods (e.g., inter-speaker supervision) suffer from overlapping features.
- **Solution**
  - **Mutual Information (MI)**: Measures dependency between variables, capturing **linear and non-linear relationships**.
  - **Estimation Approach**
    - **Challenge**: Direct MI calculation is impractical due to high dimensionality and unknown distributions.
    - **Technique**
      - Use a **variational upper bound** (Cheng et al., 2020) to approximate MI.
      - Introduce parameterized conditional distributions for efficient estimation and minimization.

Mutual Information-based Decoupling

- **Challenge**
  - Simply separating latent spaces for speaker and content embeddings does not guarantee sufficient **feature decoupling**.
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- **Solution**
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Estimation Approach

- **Challenge**: Direct MI calculation is impractical due to high dimensionality and unknown distributions.

- **Solution**
  - Use a **variational upper bound** (Cheng et al., 2020) to approximate MI.
  - Introduce parameterized conditional distributions for efficient estimation and minimization.

- **Minimizing the overlap information** between the extracted style features and content features

➢ Successfully establish a speaker style space related to identity and a content space associated with semantics.

Estimation Approach

- **Formula**

$$L_{con} = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\log\left(\frac{q\left(F_{con,i} \mid F_{spk,i}\right)}{q\left(F_{con,j} \mid F_{spk,i}\right)}\right)$$

- **Components**
  - $F_{spk,i}$: The speaker style embeddings for the i-th sample
  - $F_{con,i}, F_{con,j}$: The content embeddings for the i-th and j-th samples

$$I(F_{\text{con}}, F_{\text{spk}}) = \int p(F_{\text{con}}, F_{\text{spk}}) \log \frac{p(F_{\text{con}}, F_{\text{spk}})}{p(F_{\text{con}})p(F_{\text{spk}})}$$

원래 MI 식

Speech-related Speaker-Identity Supervision

- **Formula**

$$L_{id-s} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{i=1}^{C} t_{ni} \cdot \log(p_{ni})$$

- **Components**
  - C: The number of distinct speakers
  - t_{ni}: The one-hot encoded target label for the n-th sample
  - p_{ni}: The softmax probabilities that the n-th sample's the speaker feature F_{spk} belong to the i-th speaker

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Speech-related Speaker-Identity Supervision

- **Formula**

$$L_{id-s} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{i=1}^{C} t_{ni} \cdot \log(p_{ni})$$

- **Objective**
  - Enforce **identity consistency** for the **same speaker**.
  - Promote **identity diversity** across **different speakers**.

- These two loss functions prevent the generation of overly similar outputs.

➢ Protecting against **mode collapse**.

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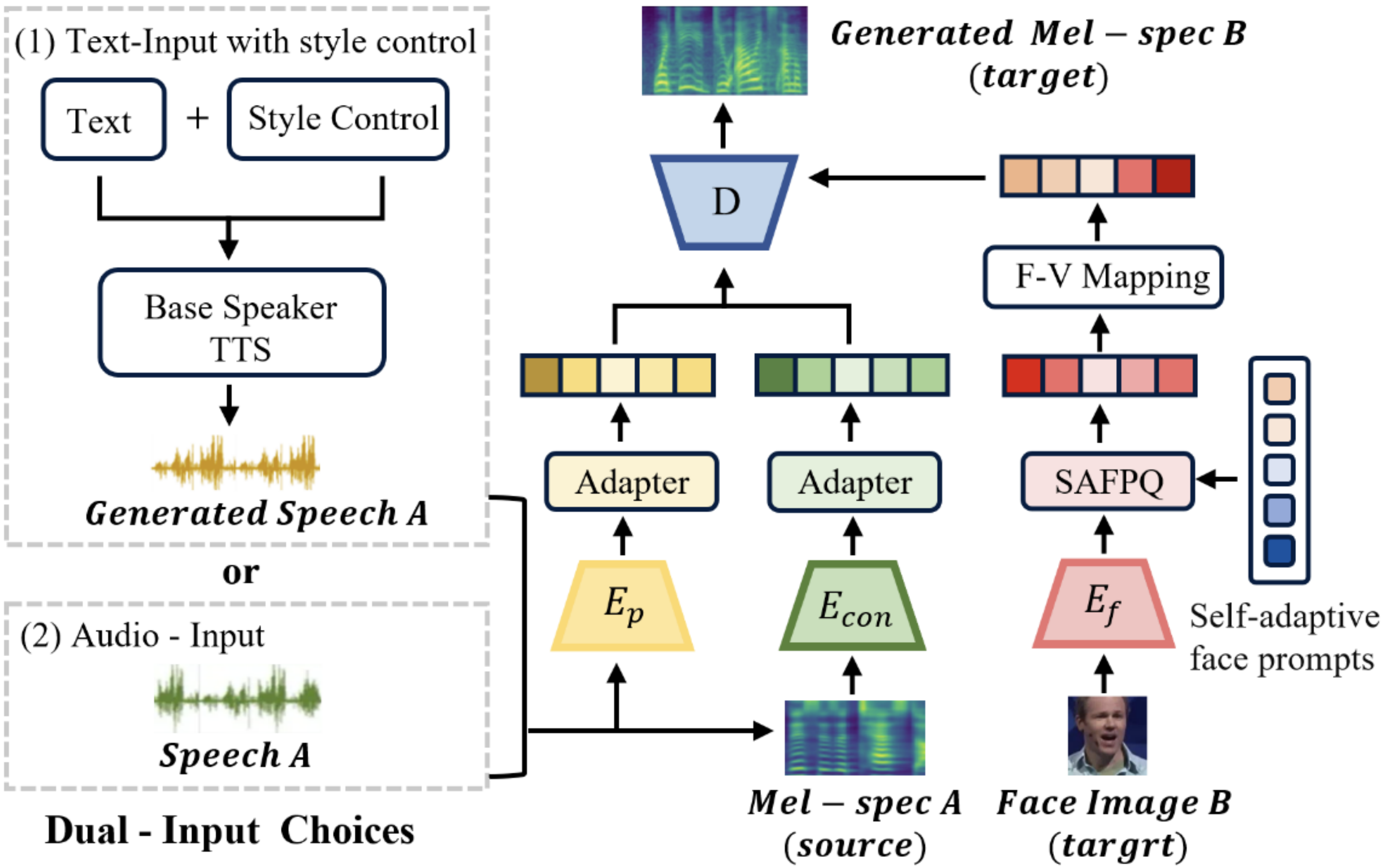
Alternative Text-Input with Style Control



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Training Loss

- **Formula**

$$L_{rec} = \left\| Mel - \widehat{Mel} \right\|_2^2$$

$$L = L_{rec} + \lambda_1 L_{con} + \lambda_2 L_{MI} + \lambda_3 L_{id-f} + \lambda_4 L_{id-s} + \lambda_5 L_F$$

- L_F: Inter-speaker Supervision Loss + Face-Voice Mapping Loss in FVMVC

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Experiment and Result

Table 2: Comparison with SOTA methods. Best performances are highlighted in bold, while second-best are underlined.

Input Modalities	Methods	Naturalness			Similarity	Consistency & Diversity	
		UTMOS ↑	WER ↓	CER ↓	SECS ↑	SEC ↑	SED ↓
Audio Input	FaceVC (Lu et al. 2021)	2.155	16.67%	10.79%	0.702	0.986	0.965
	SP-FaceVC (Weng, Shuai, and Cheng 2023)	1.831	29.17%	19.67%	**0.723**	**0.988**	0.912
	FVMVC (Sheng et al. 2023)	3.023	21.79%	15.02%	0.678	0.987	0.835
	Ours (ID-FaceVC)	**3.286**	**12.11%**	**7.86%**	0.713	**0.988**	**0.832**
Text Input	FaceTTS (Lee, Chung, and Chung 2023)	2.102	14.31%	8.70%	0.701	0.987	0.912
	Ours (ID-FaceVC)	**3.454**	**5.02%**	**2.69%**	**0.704**	**0.989**	**0.844**

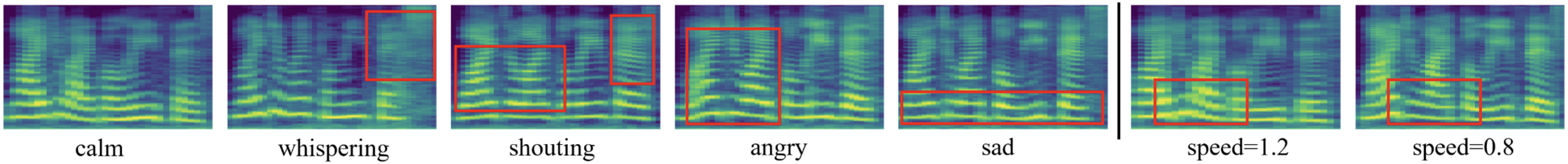

Figure 4: Mel-spectrogram visualizations of voices generated from text inputs under different emotions and speeds. The red boxes highlight areas with significant changes compared with the "calm" state.

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Experiment and Result

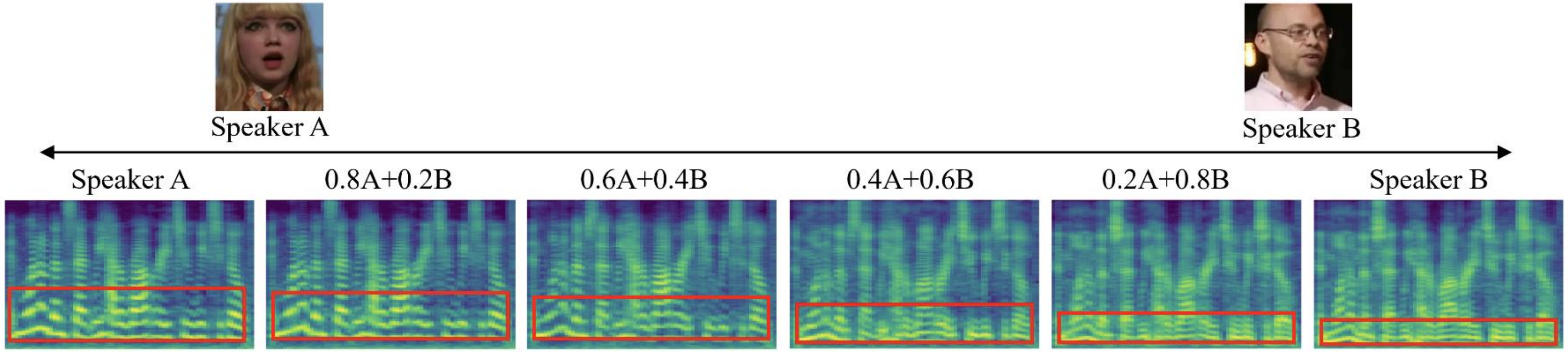

Figure 5: The mel-spectrograms of voices generated by mixed facial embeddings. From left to right, as the weight of male facial embeddings increases, the voice characteristics gradually shift from female to male, and the fundamental frequency decreases.

Experiment and Result

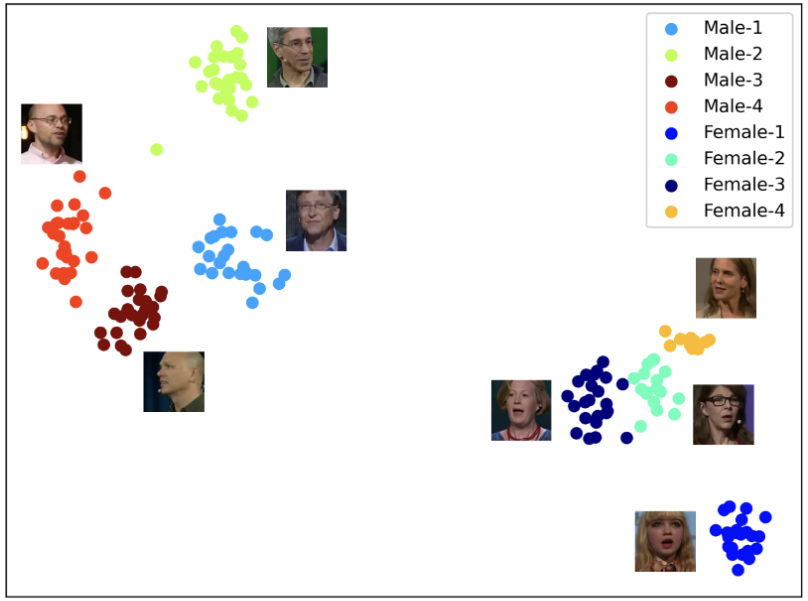

Figure 6: The t-SNE visualization of speaker embeddings form generated speech. Each point represents a voice sample, with nearby images showing the faces of the speakers.

Table 3: Results of ablation studies on different model components. Best performances are highlighted in bold, while second-best performances are underlined.

IAQ-CL	MIDD	L_{id-f}	L_{id-s}	UTMOS ↑	WER ↓	CER ↓	SECS ↑
				2.945	15.29%	10.04%	0.693
✓				3.227	18.04%	11.57%	**0.726**
✓	✓			3.266	12.70%	8.63%	0.709
✓	✓	✓		3.236	12.25%	7.97%	0.709
✓	✓		✓	3.221	**12.01%**	8.01%	0.712
✓	✓	✓	✓	**3.286**	12.11%	**7.86%**	0.713

Experiment and Result

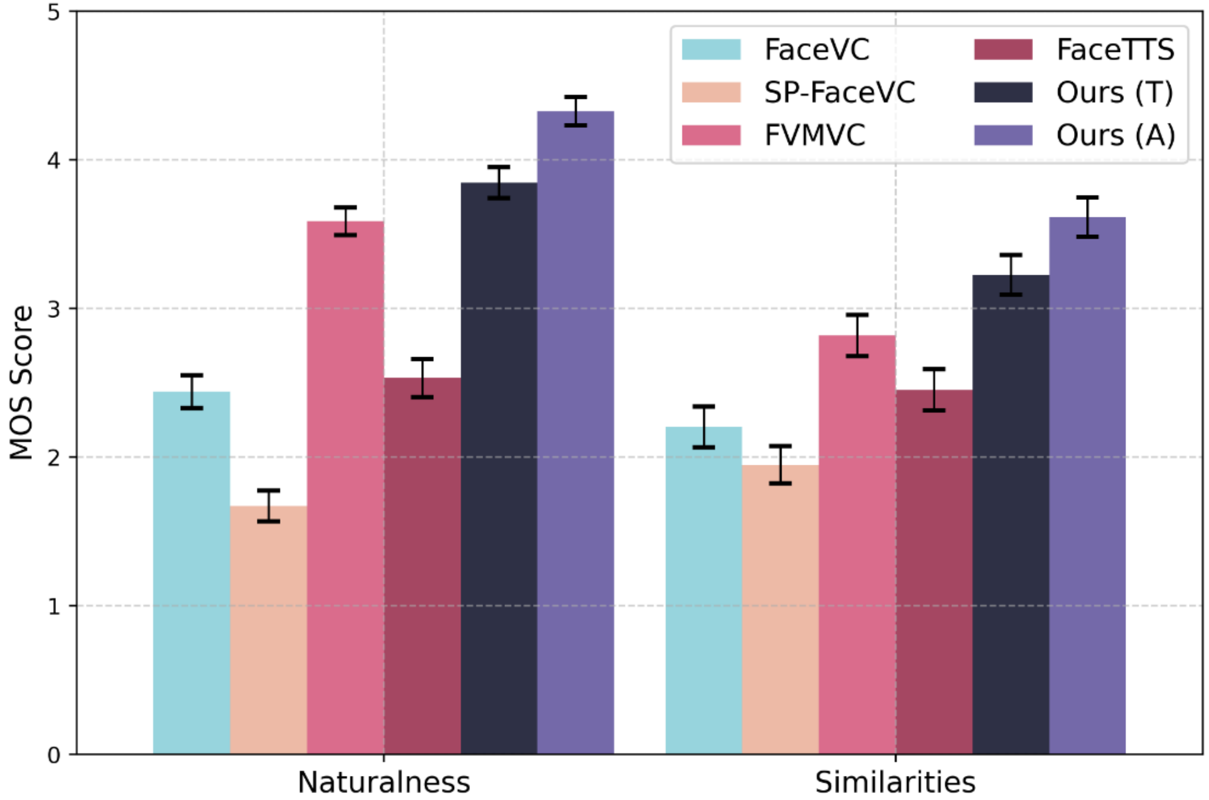

Figure 7: User study results for naturalness and similarity metrics. Ours (T) and Ours (A) represent ID-FaceVC with text and audio inputs, respectively.

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Conclusion

- **Framework**
  - Introduced the **ID-FaceVC** framework to generate speech aligned with **facial identity features**.

- **Key Components**
  - **IAQ-CL Module**: Maps facial features to speech.
  - **MIDD Module**: Enhances feature disentanglement for precise mapping.
  - **Text Modality**: Enables control over **speech content**, ensuring **naturalness, rhythm, and controllability**.

- **Future Directions**
  - Extend beyond audio generation to include **expressive facial animations**, making the output both **audible** and **visible**.


Thank You!